Training-Time Distillation for Gaze-Only Reading-Goal Classification

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***Abstract*—Reading interfaces could adapt to whether users read for comprehension or seek specific information. Prior work combines gaze and contextual text and requires fixation-to-word alignment at inference. We instead use a word-aligned teacher only during training to supervise a GRU that receives the chronological fixation sequence. On OneStop, with readers and articles both held out, distillation raises balanced accuracy by 3.39 percentage points over the same GRU trained with hard labels. The improvement remains positive under reader-clustered and crossed reader/article bootstrap analyses. Shuffling teacher targets within each class reduces the gain by 1.90 points but does not eliminate it. Adding contextual text to the word-level teacher yields no detectable student benefit. On one laptop CPU thread, median and 95th-percentile forward-pass latencies are 5.08 and 16.57 ms.**

***Index Terms*—eye tracking, scanpath analysis, reading behavior, human–computer interaction, sequence classification**

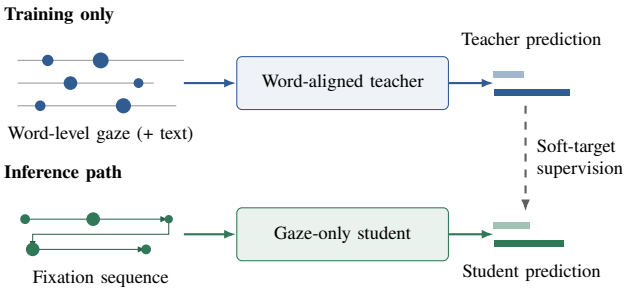


Fig. 1. Training-time distillation for gaze-only inference. A word-aligned teacher aggregates gaze by word and may also receive contextual text. Its soft predictions supervise a student that receives fixations in temporal order. Marker size represents fixation duration. At inference, the student operates without text, word alignment, or the teacher.

I. Introduction

A reader may approach the same passage to understand it broadly or to locate specific information. These goals are associated with differences in fixation durations, regressions, and scanpath structure [1], [2]. An opt-in classifier could support navigation or content assistance after each paragraph [3]. For this use, a model should generalize to readers and articles absent from training.

Shubi *et al.* combined fixation-level gaze with contextual text to classify reading goals for unseen readers and articles [3]. At inference, this model requires text, fixation-to-word alignment, and a language encoder. We ask whether word-aligned supervision can instead be used only during training while prediction uses gaze alone.

We study teachers that aggregate gaze by word. One teacher also receives contextual word representations. Their soft predictions supervise a student that observes the chronological fixation sequence. The primary comparison isolates the training signal by holding the student architecture and input fixed. At inference, the student uses no teacher, text encoder, tokenization, or fixation-to-word alignment. Fig. 1 summarizes this separation between training and inference.

Each OneStop reader completes only one goal condition, while articles recur across readers. We therefore hold out readers and articles simultaneously and estimate uncertainty by resampling readers, with crossed reader/article resampling as a sensitivity analysis.

This study makes three contributions:

- With student inputs and architecture fixed, teacher-derived soft targets improve the gaze-only GRU when both readers and articles are unseen.
- Within-class target shuffling reduces but does not eliminate the gain, while adding text to the word-level teacher provides no detectable student benefit.
- We characterize student size and single-thread CPU latency across the observed sequence-length distribution.

II. Related Work

A. Eye Movements and Reading Goals

OneStop includes eye movements from ordinary reading for comprehension, information seeking, and repeated reading [4].

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It has also supported gaze-only and text–gaze models for predicting reading comprehension under new-reader and new-item conditions [5].

Shubi *et al.* introduced this reading-goal decoding task under new-reader and new-item conditions and evaluated gaze-only and text–gaze models at fixation and word levels [3]. We use a word-level teacher to place contextual text and aggregated gaze in a common representation. This supports a controlled removal of text while the student's chronological fixation input remains fixed.

B. Distillation with Privileged Information

Knowledge distillation (KD) trains a student with class labels and a teacher's temperature-softened predictive distribution [6]. Generalized distillation allows the teacher to use privileged information that is unavailable to the student at inference [7]. This setting allows word-level gaze, with or without contextual text, to guide training while the deployed model remains gaze-only. We apply it to reading-goal classification and test whether the benefit depends on contextual text or the pairing between each training trial and its teacher prediction.

III. Method

A. Task and Data

The unit of analysis is a completed paragraph-reading trial represented by its eye movements. The binary target is the experimentally assigned reading-goal condition: ordinary reading for comprehension ($y = 0$) or information seeking ($y = 1$). In the information-seeking condition, readers preview the question before the paragraph. In ordinary reading, readers see the question only after reading the paragraph. Readers were randomly assigned to one condition [4]. We use the anonymized public release, collected with consent under an institutional ethics protocol, and retain first readings from these two conditions. We exclude repeated readings, question-screen events, answers, correctness, and all post-reading information.

The resulting data contain 19,438 trials from 360 readers and 30 articles, with nearly balanced goal labels. Each reader completed one goal condition, and the median trial contains 96 valid fixations. Reader identity, article identity, question content, session position, and condition metadata are not model inputs.

Each fixation is encoded by ten features: x and y coordinates normalized to the paragraph bounds, log fixation duration, horizontal and vertical displacement from the preceding fixation, and log saccade amplitude. The remaining features are the sine and cosine of the saccade angle, a line-transition indicator, and position within the observed sequence. Retained fixations have positive finite durations and coordinates within the paragraph region. Continuous features are standardized using statistics from the current training set only.

B. Models

Table I summarizes the five neural model variants. Models trained without distillation minimize binary cross-entropy (BCE). Gaze-summary MLP maps 22 sequence summaries to a two-hidden-layer multilayer perceptron (MLP) with widths 128 and 64, ReLU activations, and dropout 0.2. The summaries cover duration, displacement, saccade amplitude, line transitions, spatial dispersion, and path length.

TABLE I
Model configurations and trainable parameter counts. GRU rows share an architecture.

Model	Objective	Model input	Parameters
Gaze-summary MLP	BCE	gaze summaries	11,265
Gaze GRU	BCE	fixation sequence	43,797
Text–gaze teacher	BCE	text + word-level gaze	1,038,945[a]
Text-only control	BCE	text	1,038,945[a]
Distilled GRU	BCE + KD	fixation sequence	43,797

[a] Trainable parameters, excluding frozen RoBERTa.

Gaze GRU applies layer normalization to each fixation and processes the sequence with a two-layer unidirectional GRU [8] with hidden size 64. A 64-unit ReLU layer with dropout 0.2 maps the final GRU state to the output logit. This model is the matched undistilled baseline for Distilled GRU. Sequences are zero-padded within each minibatch, and the classifier uses the GRU output at the trial's final fixation before padding.

A frozen RoBERTa-base encoder produces contextual subword representations [9]. For each paragraph word, Text–gaze teacher averages its subword representations. Eight word-level gaze features encode fixation count, total and mean dwell time, first-fixation and first-pass dwell time, regression-in and revisit counts, and skipping. Text and gaze features are projected to 128 and 64 dimensions, concatenated with a 32-dimensional learned word-position embedding, and mapped to width 192. A learned classification token and a three-layer, six-head Transformer [10] with feed-forward width 384, GELU, and dropout 0.1 produce the logit. A padding mask excludes padded word positions. Text-only control retains the text branch and fusion architecture but sets all gaze features to zero. It tests whether paragraph text alone predicts the reading goal.

Distilled GRU has exactly the same architecture and inputs as Gaze GRU. Let z_S and z_T be the student and teacher logits, and let $\tau = 2$. The student minimizes

$$\mathcal{L}_D = (1-\alpha)\,\mathrm{BCE}(y, \sigma(z_S)) + \alpha\tau^2\,\mathrm{KL}\left[q_T^{(\tau)} \,\middle\|\, q_S^{(\tau)}\right], \quad q_j^{(\tau)} = \mathrm{Bern}\left(\sigma\left(\frac{z_j}{\tau}\right)\right). \tag{1}$$

In the primary analysis, Distilled GRU uses targets from Text–gaze teacher. The word-aligned gaze teacher is evaluated as a post hoc control in Sec. III-D. Each teacher produces in-sample logits for the outer-training trials on which it was fit. These logits are used only to train the student. No held-out test trial is used to fit either model or construct a distillation target.

C. Evaluation Protocol

OneStop assigns each reader to one of three article batches. Readers are divided into five folds stratified by goal and batch, and articles are divided independently into five folds. Every reader fold contains both goals from all three batches. The

Cartesian product produces disjoint test partitions that each contain both goals and together cover all retained trials once. For each reader–article fold pair, training excludes every trial from either held-out fold, and the test set is their intersection. All test readers and articles are therefore absent from training.

Within each outer training set, model selection uses three paired inner validation splits. Each validation split likewise holds out both its readers and articles from the corresponding inner training set. Learning rates are selected from $\{10^{-3}, 3\times 10^{-4}\}$ for the MLP and GRU models and $\{3\times 10^{-4}, 10^{-4}\}$ for the teacher models. We also select $\alpha \in \{0.25, 0.50, 0.75\}$ for the objective in (1). Balanced accuracy is the selection metric, with negative log-likelihood (NLL) as the first tie-breaker. Each final model is refitted for the median number of epochs selected across the three inner splits. The decision threshold is fixed at 0.5 before test evaluation. Test labels are not used for early stopping, calibration, or threshold selection.

Models are optimized with AdamW [11], weight decay 10^{-4}, and gradient clipping at 1.0. Inner training uses at most 30 epochs with a patience of five epochs. Batch sizes are 512 for Gaze-summary MLP, 256 for the GRU models, and 128 for the teacher models. Final refits use the selected learning rate and fixed epoch count described above, without test-dependent stopping.

D. Baseline and Ablation Controls

We used the same outer splits and training setup for the following post hoc controls. Paired conditions inherit the corresponding model's selected learning rate and epoch count and keep the remaining student, teacher, or KD settings fixed.

The first control replaces KD with binary label smoothing ($\epsilon = 0.10$) [12]. The second creates a word-aligned gaze teacher by zeroing every RoBERTa representation while retaining the architecture and word-level gaze summaries. Across three random seeds, we refit the BCE and label-smoothed GRUs, both teachers, and both KD students. Each student uses logits from the teacher trained with the same seed.

The third control cyclically reassigns teacher logits within each ground-truth class. This preserves each class-conditional distribution while removing the pairing between a training trial and its teacher prediction. It uses the same seeds and student settings.

We also fit logistic regression (LR) controls using either log dwell time per word [3] or the 22 gaze summaries. Their regularization is selected within the same nested reader/article splits.

E. Metrics and Uncertainty

Balanced accuracy (BA) is the primary metric because it averages the two goal-specific recalls [13]. Secondary metrics are area under the receiver operating characteristic curve (AUROC), macro F1, negative log-likelihood (NLL), Brier score, and 15-bin expected calibration error (ECE) [14].

For the primary contrast between Distilled GRU and Gaze GRU, a paired 5,000-draw percentile bootstrap resamples whole readers within each goal. We report the central 95% interval. A 2,000-draw sensitivity analysis resamples readers within each goal and independently resamples articles, weighting each trial by the product of their multiplicities. Bootstrap samples are paired between models. For each ablation contrast, resampled units are paired across conditions, and the seed-specific BA differences are averaged within each draw.

TABLE II

Out-of-fold test performance with both readers and articles held out. Lower is better for NLL, Brier, and ECE.

Model	BA	AUROC	Macro F1	NLL	Brier	ECE
Dwell-time LR[a]	.5887	.6292	.5887	.6635	.2359	.0155
Gaze-summary LR[a]	.6066	.6493	.6065	.6572	.2327	.0236
Gaze-summary MLP	.5942	.6359	.5936	.6632	.2360	.0410
Gaze GRU	.5866	.6178	.5862	.6710	.2391	.0210
Distilled GRU	**.6205**	**.6708**	**.6204**	**.6427**	**.2265**	**.0142**
Text–gaze teacher	.6126	.6613	.6120	.6561	.2312	.0496
Text-only control	.4986	.5017	.4986	.6952	.2510	.0304

[a]LR controls use nested selection within the same reader/article splits.

TABLE III

Post hoc matched-seed controls. BA is in percent. Differences and reader-clustered CIs are in percentage points (pp). Model rows compare with GRU+BCE, and n_+ counts positive seed-wise differences.

Condition	BA (%)	ΔBA (pp)	95% CI (pp)	n_+
GRU+BCE	58.57	–	–	–
GRU+label smoothing	58.52	−0.05	[−0.22, 0.11]	1/3
KD: aligned gaze teacher	62.21	+3.64	[1.99, 5.34]	3/3
KD: aligned text–gaze teacher	62.32	+3.76	[2.23, 5.29]	3/3
KD: class-shuffled targets	60.42	+1.85	[0.52, 3.17]	3/3
KD: aligned text–gaze − gaze	–	+0.11	[−0.30, 0.52]	1/3
KD: unshuffled − class-shuffled	–	+1.90	[0.61, 3.17]	3/3

F. Model-Only Latency

We benchmark one distilled checkpoint on every retained fixation sequence. After 100 warm-up iterations, we time one full-sequence forward pass per trial in randomized order. Measurements use an Intel Core Ultra 7 155H and one PyTorch thread, excluding input construction, parsing, and disk I/O.

IV. Results

A. Primary Performance

Table II reports pooled out-of-fold predictions from the prespecified primary run, which used one fixed initialization. The seed analysis in Table III averages three matched initializations.

Distilled GRU raises BA by 3.39 percentage points relative to Gaze GRU. Reader-clustered resampling gives a 95% interval of $[1.75, 5.04]$ points, and crossed reader/article resampling gives $[1.30, 5.53]$ points. Its NLL, Brier score, and ECE are also lower than those of the matched GRU. Text-only control is near chance.

Gaze-summary LR reaches 0.6066 BA. Distilled GRU is 1.39 percentage points higher, but its crossed reader/article interval includes zero ($[-0.37, 3.21]$ points).

B. Ablations and Seed Sensitivity

Table III reports the post hoc controls.

The controls separate the tested training signals. The label-smoothed GRU remains at the BCE baseline. Both word-aligned teachers yield positive seed-wise KD differences, while the student-level modality contrast includes zero under both resampling schemes. Within-class shuffling preserves the class-conditional target distribution but removes trial-level pairing. The unshuffled-versus-shuffled contrast excludes zero under both schemes, although shuffled targets remain above BCE.

C. Model-Only Latency

The distilled and undistilled GRUs have identical parameter counts and inference operations. The serialized student state occupies 176 KiB. Across all retained sequences, model-only latency has a 5.08-ms median and a 16.57-ms 95th percentile.

V. Discussion

Teacher-derived soft targets improve the matched gaze-only GRU relative to hard-label BCE. The gaze-summary logistic baseline remains competitive, and the distilled model's advantage over it is uncertain under crossed reader/article resampling.

The ablations suggest that the gain has more than one source. Trial-to-target pairing contributes because shuffling within class reduces performance. The remaining improvement may reflect class-dependent targets or an optimization effect. The tested label-smoothing setting does not yield a similar improvement, and adding contextual text produces no detectable student benefit. Both teachers also share word-level gaze aggregation, greater capacity than the student, and in-sample target generation. These properties remain possible sources of the gain.

Bootstrap intervals characterize sampling uncertainty for fixed out-of-fold predictions, while the seed analysis covers three initializations. Because each reader was assigned one goal, the study does not test within-reader goal changes. Generalization beyond complete scanpaths from English news read with a laboratory eye tracker remains open.

The gaze-only student still requires calibrated coordinates and paragraph bounds. The CPU benchmark measures model-only forward passes. Adaptive interfaces should be opt-in and evaluated with users.

VI. Conclusion

Training-time distillation improves a gaze-only GRU for unseen readers and articles while inference uses only fixation sequences. Within-class target shuffling reduces the gain, whereas adding contextual text to the tested word-level teacher yields no detectable benefit. Together, these results support word-aligned teacher supervision as a training-time method for improving a fixation-sequence GRU without changing its inference inputs or operations.

Code Availability

Code, configurations, and aggregate results are available at github.com/gohyunsu/gaze-goal.

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